

Circuit Documentation

Summary of the Circuit

This circuit is designed to interface a Raspberry Pi Zero 2 W with multiple pushbuttons, an RGB LED, a buzzer, and resistors. The Raspberry Pi acts as the central control unit, receiving inputs from the pushbuttons and controlling the RGB LED and buzzer. The circuit is powered through the Raspberry Pi, and all components are interconnected to perform specific functions based on the GPIO pins of the Raspberry Pi.

Component List

1. **Red Cap Pushbutton**
 - Description: A pushbutton with a red cap used for user input.
 - Pins: 2, 4, 1, 3
2. **Pushbutton (1)**
 - Description: A standard pushbutton used for user input.
 - Pins: Pin 2, Pin 1, Pin 3, Pin 4
3. **Pushbutton (2)**
 - Description: A standard pushbutton used for user input.
 - Pins: Pin 2, Pin 1, Pin 3, Pin 4
4. **Pushbutton (3)**
 - Description: A standard pushbutton used for user input.
 - Pins: Pin 2, Pin 1, Pin 3, Pin 4
5. **Raspberry Pi Zero 2 W**
 - Description: A small, affordable computer used as the central control unit.
 - Pins: Multiple GPIO pins, power, and ground connections
6. **Buzzer (2)**
 - Description: An electronic component that produces sound.
 - Pins: VCC, GND
7. **RGB LED (Wokwi compatible)**
 - Description: An LED capable of displaying multiple colors.
 - Pins: R, COM, G, B
8. **Resistor**
 - Description: A resistor with a resistance of 220 Ohms.
 - Pins: pin1, pin2
9. **Resistor (1)**
 - Description: A resistor with a resistance of 220 Ohms.
 - Pins: pin1, pin2
10. **Resistor (2)**
 - Description: A resistor with a resistance of 220 Ohms.
 - Pins: pin1, pin2
11. **GND**
 - Description: Ground connection for the circuit.
 - Pins: GND

Wiring Details

Red Cap Pushbutton

- Pin 3 is connected to **GPIO 17 Chip Enable CE1** on the Raspberry Pi Zero 2 W.
- Pin 2 is connected to the common ground.

Pushbutton (1)

- Pin 4 is connected to **GPIO 22** on the Raspberry Pi Zero 2 W.
- Pin 1 is connected to the common ground.

Pushbutton (2)

- Pin 4 is connected to **GPIO 27** on the Raspberry Pi Zero 2 W.
- Pin 1 is connected to the common ground.

Pushbutton (3)

- Pin 4 is connected to **GPIO 23** on the Raspberry Pi Zero 2 W.
- Pin 1 is connected to the common ground.

Raspberry Pi Zero 2 W

- **GPIO 18 Chip Enable CE1 (PWM)** is connected to **VCC** of the Buzzer.
- **GPIO 6** is connected to **pin2** of a Resistor.
- **GPIO 5** is connected to **pin2** of another Resistor.
- **GPIO 13 (PWM)** is connected to **pin2** of another Resistor.
- **Ground** is connected to the common ground, which includes all pushbuttons, the RGB LED's COM pin, and the Buzzer's GND pin.

Buzzer (2)

- **VCC** is connected to **GPIO 18 Chip Enable CE1 (PWM)** on the Raspberry Pi Zero 2 W.
- **GND** is connected to the common ground.

RGB LED (Wokwi compatible)

- **R** is connected to **pin1** of a Resistor.
- **G** is connected to **pin1** of another Resistor.
- **B** is connected to **pin1** of another Resistor.
- **COM** is connected to the common ground.

Resistors

- Each resistor is connected between a GPIO pin on the Raspberry Pi and a pin on the

RGB LED.

GND

- Provides a common ground connection for the circuit.

Documented Code

There is no embedded code provided for this circuit. The functionality of the circuit is determined by the hardware connections and any external software running on the Raspberry Pi Zero 2 W.